

Fundamental Theorem of Arithmetic in $\mathbb{Q}[x]$

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1. Introduction

Theorem 1. *Theorems are massive results.*

You can also hyperlink to Theorem [1](#) if you want!

Corollary 2. *Corollaries follow from theorems.*

Remember to follow the good writing guidelines! You can also break pages if you feel it appropriate.

2. History

Historical background is often nice.

3. Axioms

You can make subsections if you find it appropriate.

3.1. Integers

Lemma 3 (NIBZO). *There is no integer strictly between 0 and 1. Equivalently, if $n \in \mathbb{Z}$ and*

$$0 < n < 1,$$

then such an n cannot exist.

Proof. Suppose, for contradiction, that there is an integer strictly between 0 and 1. Put

$$S = \{n \in \mathbb{Z} : 0 < n < 1\}.$$

Then S is nonempty and $S \subseteq \mathbb{Z}_{\geq 0}$. By the well-ordering principle, S has a least element. Call it m .

Since $0 < m$, multiplicative closure of positive integers gives $0 < m^2$. Since $m < 1$ and $m > 0$, multiplying the inequality $m < 1$ by the positive integer m gives

$$m^2 < m.$$

Thus m^2 is also an integer satisfying

$$0 < m^2 < m < 1.$$

So $m^2 \in S$, but $m^2 < m$. This contradicts the choice of m as the least element of S . Therefore S is empty. $\square$

3.2. Rationals

Lemma 4 (No zero divisors in $\mathbb{Q}$). *If $a, b \in \mathbb{Q}$ and $ab = 0$, then $a = 0$ or $b = 0$.*

Proof. Suppose $ab = 0$. If $a = 0$, there is nothing to prove. Now assume $a \neq 0$. Then a^{-1} exists in

$\mathbb{Q}$. Multiplying both sides of $ab = 0$ by a^{-1} gives

$$a^{-1}(ab) = a^{-1} \cdot 0.$$

By associativity, the left side is

$$(a^{-1}a)b = 1b = b.$$

Also $a^{-1} \cdot 0 = 0$: indeed,

$$a^{-1} \cdot 0 = a^{-1}(0 + 0) = a^{-1} \cdot 0 + a^{-1} \cdot 0,$$

and adding the additive inverse of $a^{-1} \cdot 0$ to both sides gives $a^{-1} \cdot 0 = 0$. Hence $b = 0$. Therefore $a = 0$ or $b = 0$. $\square$

3.3. The Ring $\mathbb{Q}[x]$

Definition 1 (The polynomial ring $\mathbb{Q}[x]$). We define a polynomial $f(x) \in \mathbb{Q}[x]$ with degree n if it can be written as

$$f(x) = \sum_0^n a_i x^i$$

where $n \in \mathbb{Z}_{\geq 0}$ and each coefficient a_i lies in $\mathbb{Q}$ and $a_n \neq 0$.

Theorem 5 ($\mathbb{Q}[x]$ is a commutative ring.).

Proof. We check the ring conditions. Addition stays inside $\mathbb{Q}[x]$ because each coefficient $a_i + b_i$ is rational. Multiplication also stays inside $\mathbb{Q}[x]$ because each coefficient c_k is built from finitely many sums and products of rational numbers.

The zero polynomial is the additive identity because adding zero to every coefficient changes nothing. The polynomial $-f(x)$ is the additive inverse of $f(x)$ because each coefficient becomes $a_i + (-a_i) = 0$. The constant polynomial 1 is the multiplicative identity because multiplying by 1 leaves the coefficients unchanged.

The associative and commutative laws come from the same laws in $\mathbb{Q}$, coefficient by coefficient. For instance, addition is associative because

$$(a_k + b_k) + c_k = a_k + (b_k + c_k)$$

for each coefficient. The multiplication laws and distributivity are checked in the same way from the rational-number laws.

Finally, the zero polynomial has constant coefficient 0, while the constant polynomial 1 has constant coefficient 1. Since $0 \neq 1$ in $\mathbb{Q}$, have $0_{\mathbb{Q}[x]} \neq 1_{\mathbb{Q}[x]}$. $\square$

Definition 2. For $f(x) \in \mathbb{Q}[x]$, let the **norm**, or $\|f(x)\|$ be equal to

$$\|f(x)\| = 2^{\deg f(x)}$$

with the following properties:

- $\|f(x)\| \cdot \|g(x)\| = \|f(x)g(x)\|$

Proof.

- For any $c(x) \in \mathbb{Q}[x]^\times$, $\|c(x)\| = 1$.

Proof. Since $\deg c(x) = 0$, $\|c(x)\| = 2^0 = 1$.

Lemma 6 (Multiplication). *For multiplication, if*

$$f(x) = a_0 + a_1x + \cdots + a_mx^m \quad \text{and} \quad g(x) = b_0 + b_1x + \cdots + b_nx^n,$$

then

$$f(x)g(x) = c_0 + c_1x + \cdots + c_{m+n}x^{m+n},$$

where each coefficient c_k is found by adding all products a_ib_j for which the exponents satisfy $i+j = k$.

For example,

$$c_0 = a_0b_0, \quad c_1 = a_0b_1 + a_1b_0, \quad c_{m+n} = a_mb_n.$$

Definition 3 (Zero, one, and negatives). The zero element of $\mathbb{Q}[x]$ is the polynomial $0_{Qx} = 0$. Define the degree of 0_{Qx} to be $-\infty$. The one element of $\mathbb{Q}[x]$ is the constant polynomial

$$1_{\mathbb{Q}[x]} = 1$$

If

$$f(x) = \sum_0^n a_ix^i$$

then

$$-f(x) = \sum_0^n -a_ix^i$$

Theorem 7 ($\mathbb{Q}[x]$ has no zero divisors). *If $f(x), g(x) \in \mathbb{Q}[x]$ and*

$$f(x)g(x) = 0,$$

then $f(x) = 0$ or $g(x) = 0$.

Proof. Suppose $f \neq 0$ and $g \neq 0$. Then

$$\deg(fg) = \deg(f) + \deg(g) \geq 0,$$

so fg is nonzero. This contradicts $fg = 0$. Hence at least one of f and g must be zero. $\square$

Corollary 8 (Cancellation). *If $h(x) \neq 0$ and*

$$h(x)f(x) = h(x)g(x),$$

then $f(x) = g(x)$.

Proof. Subtract the right side from the left:

$$h(f - g) = 0.$$

Since $h \neq 0$ and $\mathbb{Q}[x]$ has no zero divisors, we must have $f - g = 0$. Thus $f = g$. $\square$

4. Basic Arithmetic Facts

Lemma 9 (Uniqueness of zero and one). *The additive identity and multiplicative identity in $\mathbb{Q}[x]$ are unique. In particular, if $f(x) + z(x) = f(x)$ for every $f(x) \in \mathbb{Q}[x]$, then $z(x) = 0$. Similarly, if $f(x)u(x) = f(x)$ for every $f(x) \in \mathbb{Q}[x]$, then $u(x) = 1$.*

Proof. Suppose z acts like an additive identity for f , so $f + z = f$. Add $-f$ to both sides:

$$(-f) + (f + z) = (-f) + f.$$

By associativity and the definition of additive inverse, this becomes

$$0 + z = 0.$$

Since 0 is the additive identity, $z = 0$.

Now suppose u acts like a multiplicative identity for every polynomial. Test this on the constant polynomial 1 :

$$1 \cdot u = 1.$$

But $1 \cdot u = u$. Hence $u = 1$. $\square$

Lemma 10 (Uniqueness of additive inverses). *If $u(x)$ and $v(x)$ are both additive inverses of $f(x)$, then $u(x) = v(x)$.*

Proof. Assume

$$f + u = 0 \quad \text{and} \quad f + v = 0.$$

Then $f + u = f + v$. Add $-f$ to both sides:

$$(-f) + (f + u) = (-f) + (f + v).$$

By associativity,

$$((-f) + f) + u = ((-f) + f) + v.$$

Thus

$$0 + u = 0 + v,$$

so $u = v$. □

Lemma 11 (Zero and negatives). *For all $f(x), g(x) \in \mathbb{Q}[x]$,*

$$f(x) \cdot 0 = 0, \quad -(-f(x)) = f(x), \quad (-f(x))(-g(x)) = f(x)g(x).$$

Proof. First,

$$f \cdot 0 = f(0 + 0) = f \cdot 0 + f \cdot 0.$$

Adding the additive inverse of $f \cdot 0$ to both sides gives $f \cdot 0 = 0$.

Next, since

$$(-f) + f = 0,$$

the polynomial f is an additive inverse of $-f$. By uniqueness of additive inverses,

$$-(-f) = f.$$

For the product of negatives, first observe that

$$fg + (-f)g = (f + (-f))g = 0 \cdot g = 0.$$

Therefore $(-f)g = -(fg)$. Applying this with g replaced by $-g$ gives

$$(-f)(-g) = -(f(-g)).$$

Also,

$$fg + f(-g) = f(g + (-g)) = f \cdot 0 = 0,$$

so $f(-g) = -(fg)$. Hence

$$(-f)(-g) = -(-(fg)) = fg.$$

□

5. Divisibility

Definition 4 (Divisibility). For $a(x), b(x) \in \mathbb{Q}[x]$, we say $a(x)$ divides $b(x)$, written

$$a(x) \mid b(x),$$

if there exists $q(x) \in \mathbb{Q}[x]$ such that

$$b(x) = a(x)q(x).$$

6. The Division Algorithm in $\mathbb{Q}[x]$

For $\mathbb{Q}[x]$, degree plays the role that size plays in integer division. Since degrees are nonnegative integers, we can use the well-ordering principle to pick a remainder of smallest degree.

Theorem 12 (Division algorithm in $\mathbb{Q}[x]$). *Let $f(x), g(x) \in \mathbb{Q}[x]$ with $g(x) \neq 0$. Then there exist unique polynomials $q(x), r(x) \in \mathbb{Q}[x]$ such that*

$$f(x) = q(x)g(x) + r(x),$$

where either

$$r(x) = 0$$

or

$$\deg(r) < \deg(g).$$

Proof. Existence. Look at all possible remainders after subtracting multiples of g from f :

$$S = \{ f(x) - q(x)g(x) : q(x) \in \mathbb{Q}[x] \}.$$

If $0 \in S$, then $f = qg$ for some q , so

$$f = qg + 0,$$

and the result holds with $r = 0$.

Now suppose $0 \notin S$. Then every element of S is nonzero, so each element has a degree in $\mathbb{Z}_{\geq 0}$. Define

$$D = \{ \deg(s) : s \in S \} \subseteq \mathbb{Z}_{\geq 0}.$$

The set D is nonempty because $f - 0g = f$ lies in S . By the well-ordering principle, D has a smallest element. Choose $r \in S$ with minimum degree. Since $r \in S$, there exists $q \in \mathbb{Q}[x]$ such that

$$r = f - qg.$$

Equivalently,

$$f = qg + r.$$

It remains to show that $\deg(r) < \deg(g)$. Suppose instead that

$$\deg(r) \geq \deg(g).$$

Write

$$r(x) = a_m x^m + \cdots \quad \text{and} \quad g(x) = b_n x^n + \cdots,$$

where $a_m, b_n \neq 0$ and $m \geq n$. Since $b_n \neq 0$ and $b_n \in \mathbb{Q}$, the rational number a_m/b_n exists. Define

$$r^*(x) = r(x) - \frac{a_m}{b_n} x^{m-n} g(x).$$

The second term has leading term $a_m x^m$, so the leading term of r cancels. Thus either $r^* = 0$ or

$$\deg(r^*) < \deg(r).$$

Also,

$$\begin{aligned} r^* &= f - qg - \frac{a_m}{b_n} x^{m-n} g \\ &= f - \left(q + \frac{a_m}{b_n} x^{m-n} \right) g. \end{aligned}$$

Therefore $r^* \in S$. If $r^* = 0$, then $0 \in S$, which is not the case we are in. If $r^* \neq 0$, then r^* is in S and has smaller degree than r , contradicting how we chose r . Therefore our assumption was false, and

$$\deg(r) < \deg(g).$$

Uniqueness. Suppose

$$f = q_1 g + r_1 = q_2 g + r_2,$$

where r_1 and r_2 are both either zero or have degree less than $\deg(g)$. Subtracting gives

$$g(q_1 - q_2) = r_2 - r_1.$$

Assume $q_1 \neq q_2$. Then $q_1 - q_2 \neq 0$, and since $\mathbb{Q}[x]$ has no zero divisors,

$$g(q_1 - q_2) \neq 0.$$

Hence

$$\deg(g(q_1 - q_2)) = \deg(g) + \deg(q_1 - q_2) \geq \deg(g).$$

But $r_2 - r_1$ is either zero or has degree strictly less than $\deg(g)$, since both r_1 and r_2 have degree less than $\deg(g)$. This is impossible. Therefore $q_1 = q_2$. Substituting back gives $r_1 = r_2$. $\square$

7. Math

8. Notation

Definition 5. Define **summation**, or $\sum_{i=i_0}^n a_i$ for $n \in \mathbb{Z}^+$ and $n \geq i_0$ as follows:

$$\sum_{i=i_0}^{i_0} = a_{i_0} \quad \sum_{i=i_0}^n a_i = a_n + \sum_{i=i_0}^{n-1} a_i$$

Lemma 13. $\sum_{i=i_0}^n a_i + \sum_{i=i_0}^n b_i = \sum_{i=i_0}^n (a_i + b_i)$

Proof. Note that

$$\sum_{i=i_0}^{i_0} a_i + \sum_{i=i_0}^{i_0} b_i = a_{i_0} + b_{i_0} = \sum_{i=i_0}^{i_0} (a_i + b_i)$$

Let $S = \left\{ k \in \mathbb{Z}^+ \mid \sum_{i=i_0}^{i_0+k} a_i + \sum_{i=i_0}^{i_0+k} b_i \neq \sum_{i=i_0}^{i_0+k} (a_i + b_i) \right\}$. For the sake of contradiction, S is not empty.

As $S \subseteq \mathbb{Z}^+$, by the Well-Ordering Principle, there exists a least element $l \in S$.

If $k = 1$:

$$\sum_{i=i_0}^{i_0+1} a_i + \sum_{i=i_0}^{i_0+1} b_i = a_{i_0+1} + a_{i_0} + b_{i_0+1} + b_{i_0} = (a_{i_0+1} + b_{i_0+1}) + (a_{i_0} + b_{i_0}) = \sum_{i=i_0}^{i_0+1} (a_i + b_i)$$

Thus, $1 \notin S$ and so $l > 1$ by NIBZO. Since $l - 1 \in \mathbb{Z}$ yet $l - 1 \notin S$,

$$\begin{aligned} \sum_{i=i_0}^{i_0+l-1} a_i + \sum_{i=i_0}^{i_0+l-1} b_i &= \sum_{i=i_0}^{i_0+l-1} (a_i + b_i) \\ a_l + b_l + \sum_{i=i_0}^{i_0+l-1} a_i + \sum_{i=i_0}^{i_0+l-1} b_i &= a_l + b_l + \sum_{i=i_0}^{i_0+l-1} (a_i + b_i) \\ a_l + \sum_{i=i_0}^{i_0+l-1} a_i + b_l + \sum_{i=i_0}^{i_0+l-1} b_i &= (a_l + b_l) + \sum_{i=i_0}^{i_0+l-1} (a_i + b_i) \\ \sum_{i=i_0}^{i_0+l} a_i + \sum_{i=i_0}^{i_0+l} b_i &= \sum_{i=i_0}^{i_0+l} (a_i + b_i) \end{aligned}$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. $\square$

Lemma 14. $\sum_{i=i_0}^n ca_i = c \sum_{i=i_0}^n a_i$ *Proof.* Note that

$$\sum_{i=i_0}^{i_0} ca_i = ca_{i_0} = c \sum_{i=i_0}^{i_0} a_i$$

and that

$$\sum_{i=i_0}^{i_0+1} ca_i = ca_{i_0} + ca_{i_0+1} = c \sum_{i=i_0}^{i_0+1} a_i$$

Let $S = \left\{ k \in \mathbb{Z}^+ \mid \sum_{i=i_0}^{i_0+k} ca_i \neq c \sum_{i=i_0}^{i_0+k} a_i \right\}$. For the sake of contradiction, S is not empty. As $S \subseteq \mathbb{Z}^+$, by the well-ordering principle, there exists a least element $l \in S$. Since $1 \notin S$, $l > 1$ by NIBZO. As $l-1 \in \mathbb{Z}$ yet $l-1 \notin S$,

$$\begin{aligned} \sum_{i=i_0}^{i_0+l-1} ca_i &= c \sum_{i=i_0}^{i_0+l-1} a_i \\ ca_l + \sum_{i=i_0}^{i_0+l-1} ca_i &= ca_l + c \sum_{i=i_0}^{i_0+l-1} a_i \\ \sum_{i=i_0}^{i_0+l} ca_i &= c \sum_{i=i_0}^{i_0+l} a_i \end{aligned}$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. $\square$

Definition 6. Define **product**, or $\prod_{i=i_0}^n a_i$ for $n \in \mathbb{Z}^+$ and $n \geq i_0$ as follows:

$$\prod_{i=i_0}^{i_0} a_i = a_{i_0} \quad \prod_{i=i_0}^n a_i = a_n \cdot \prod_{i=i_0}^{n-1} a_i$$

Lemma 15. $\prod_{i=i_0}^n a_i \prod_{i=i_0}^n b_i = \prod_{i=i_0}^n a_i b_i$

Proof. Note that

$$\prod_{i=i_0}^{i_0} a_i \prod_{i=i_0}^{i_0} b_i = a_{i_0} b_{i_0} = \prod_{i=i_0}^{i_0} a_i b_i$$

Let $S = \left\{ k \in \mathbb{Z}^+ \mid \prod_{i=i_0}^{i_0+k} a_i \prod_{i=i_0}^{i_0+k} b_i \neq \prod_{i=i_0}^{i_0+k} a_i b_i \right\}$. For the sake of contradiction, S is not empty. As $S \subseteq \mathbb{Z}^+$, by the Well-Ordering Principle, there exists a least element $l \in S$.

If $k = 1$:

$$\prod_{i=i_0}^{i_0+1} a_i \prod_{i=i_0}^{i_0+1} b_i = a_{i_0+1} a_{i_0} \cdot b_{i_0+1} b_{i_0} = (a_{i_0+1} b_{i_0+1}) (a_{i_0} b_{i_0}) = \prod_{i=i_0}^{i_0+1} a_i b_i$$

Thus, $1 \notin S$ and so $l > 1$ by NIBZO. Since $l - 1 \in \mathbb{Z}$ yet $l - 1 \notin S$,

$$\begin{aligned} \prod_{i=i_0}^{i_0+l-1} a_i \prod_{i=i_0}^{i_0+l-1} b_i &= \prod_{i=i_0}^{i_0+l-1} a_i b_i \\ a_l b_l \cdot \prod_{i=i_0}^{i_0+l-1} a_i \prod_{i=i_0}^{i_0+l-1} b_i &= a_l b_l \cdot \prod_{i=i_0}^{i_0+l-1} a_i b_i \\ (a_l \prod_{i=i_0}^{i_0+l-1} a_i) \cdot (b_l \prod_{i=i_0}^{i_0+l-1} b_i) &= (a_l b_l) \prod_{i=i_0}^{i_0+l-1} a_i b_i \\ \prod_{i=i_0}^{i_0+l} a_i \prod_{i=i_0}^{i_0+l} b_i &= \prod_{i=i_0}^{i_0+l} a_i b_i \end{aligned}$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. $\square$

9. Conclusion

Conclude with the big ideas!